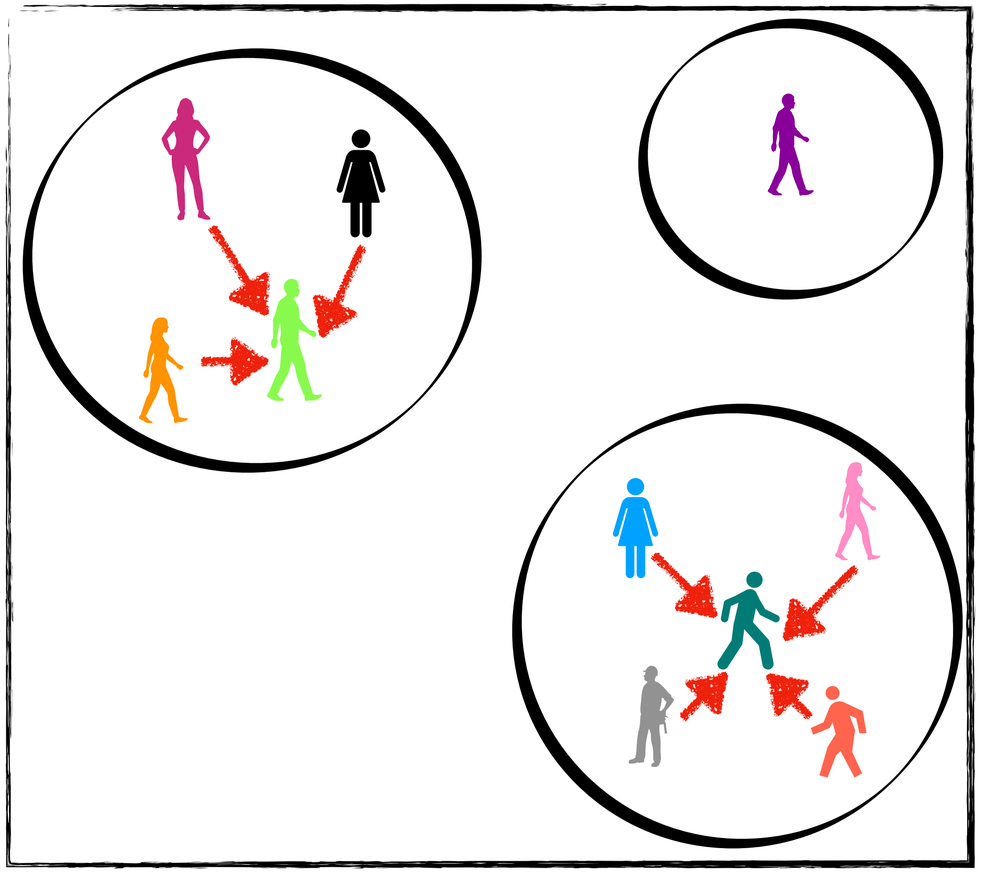
# Clustering with fair centre representation:
## parameterised approximation algorithms and heuristics

**Suhas Thejaswi**[1], Ameet Gadekar[1], Bruno Ordozgoiti[2], Michal Osadnik[1]

[1] Department of Computer Science,
Aalto University, Finland

[2] School of Electronic Engineering and Computer Science,
Queen Mary University of London, UK

# Motivation

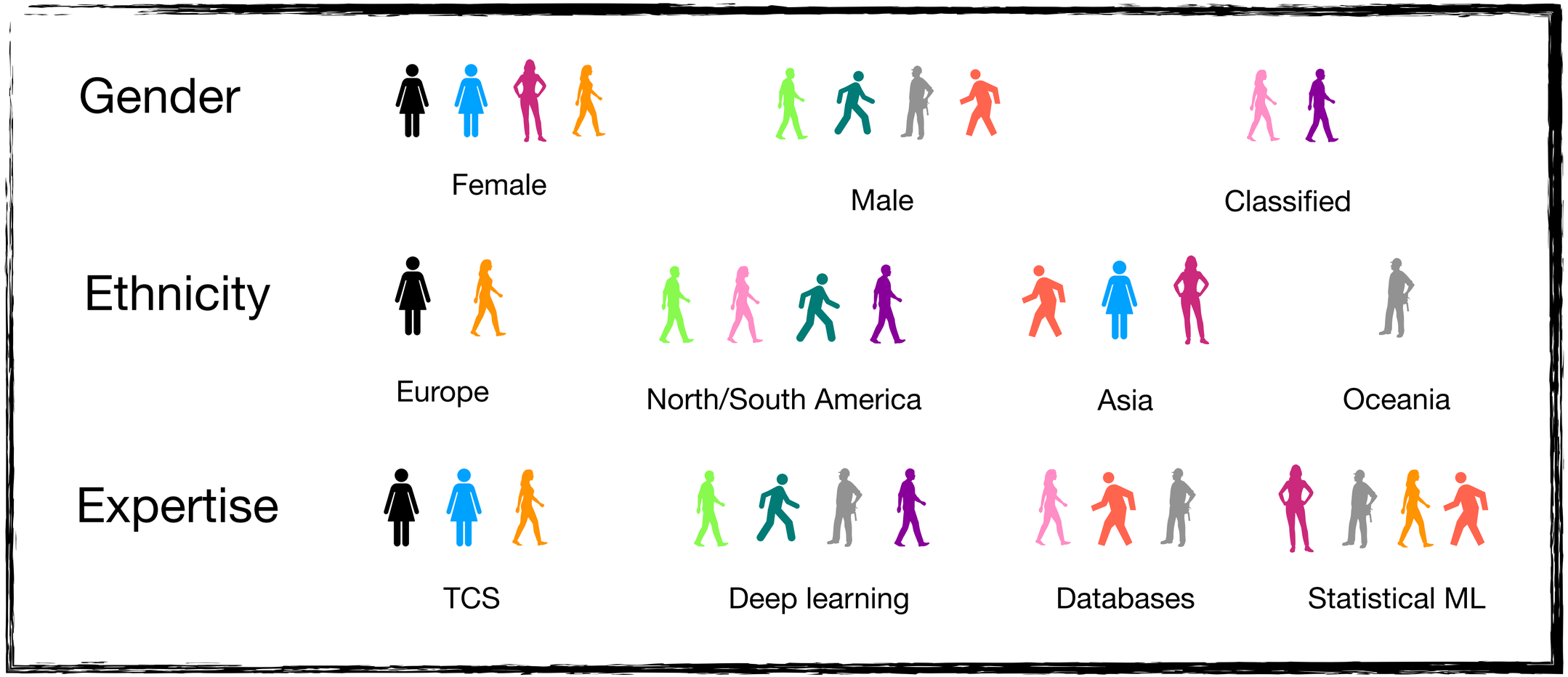


- **task**: form a committee of three representatives over a set of individuals
- **distance** — value measuring the strength of ties between individuals, for example, agreement over a set of issues
- **possible solution**: form three clusters that reduces the total distance between selected representatives (cluster centres) and individuals

**chosen committee:**

**Is the solution diverse?**

# Motivation


Gender
Female
Male
Classified
Ethnicity
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North/South America
Asia
Oceania
Expertise
TCS
Deep learning
Databases
Statistical ML

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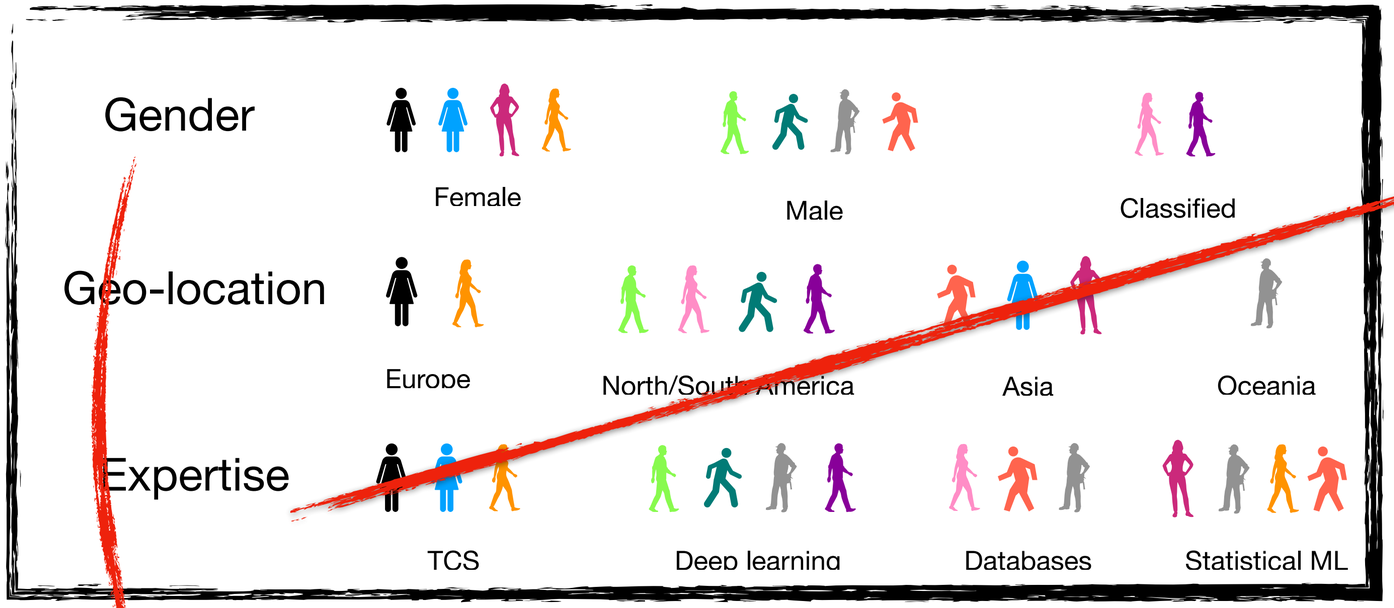
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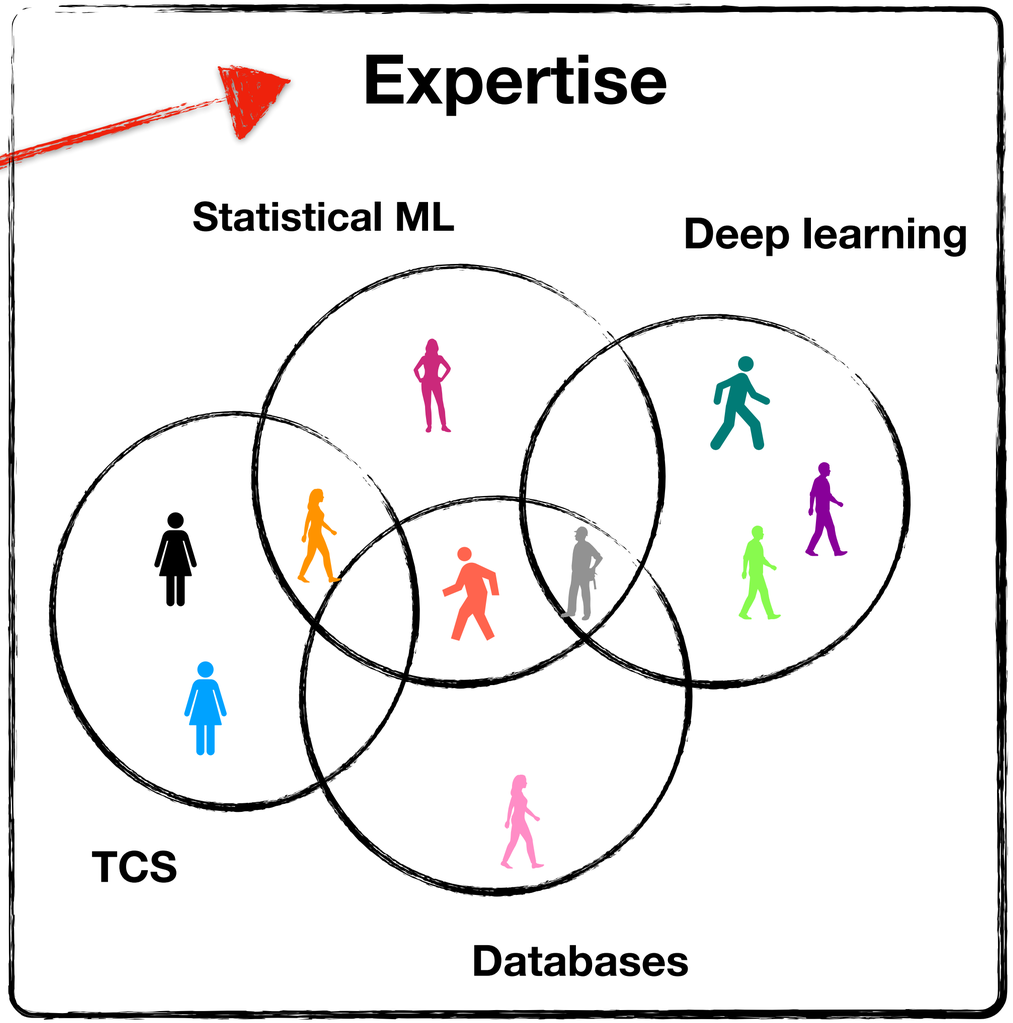
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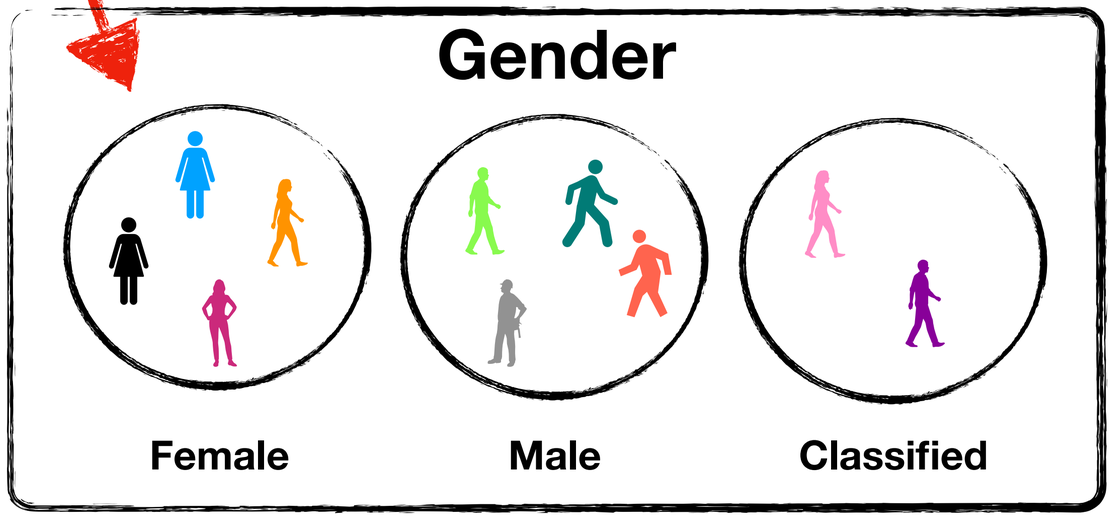
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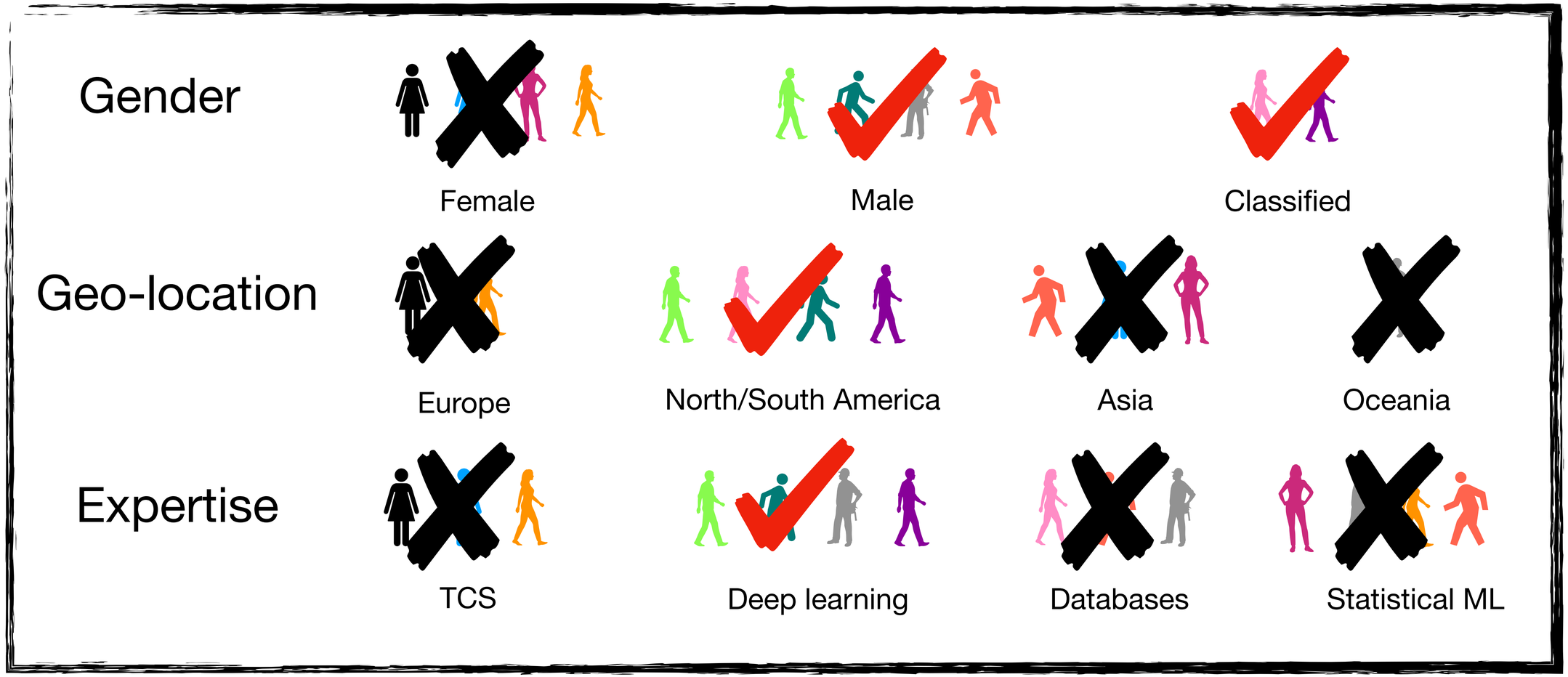



**chosen solution:** **(naive clustering)**

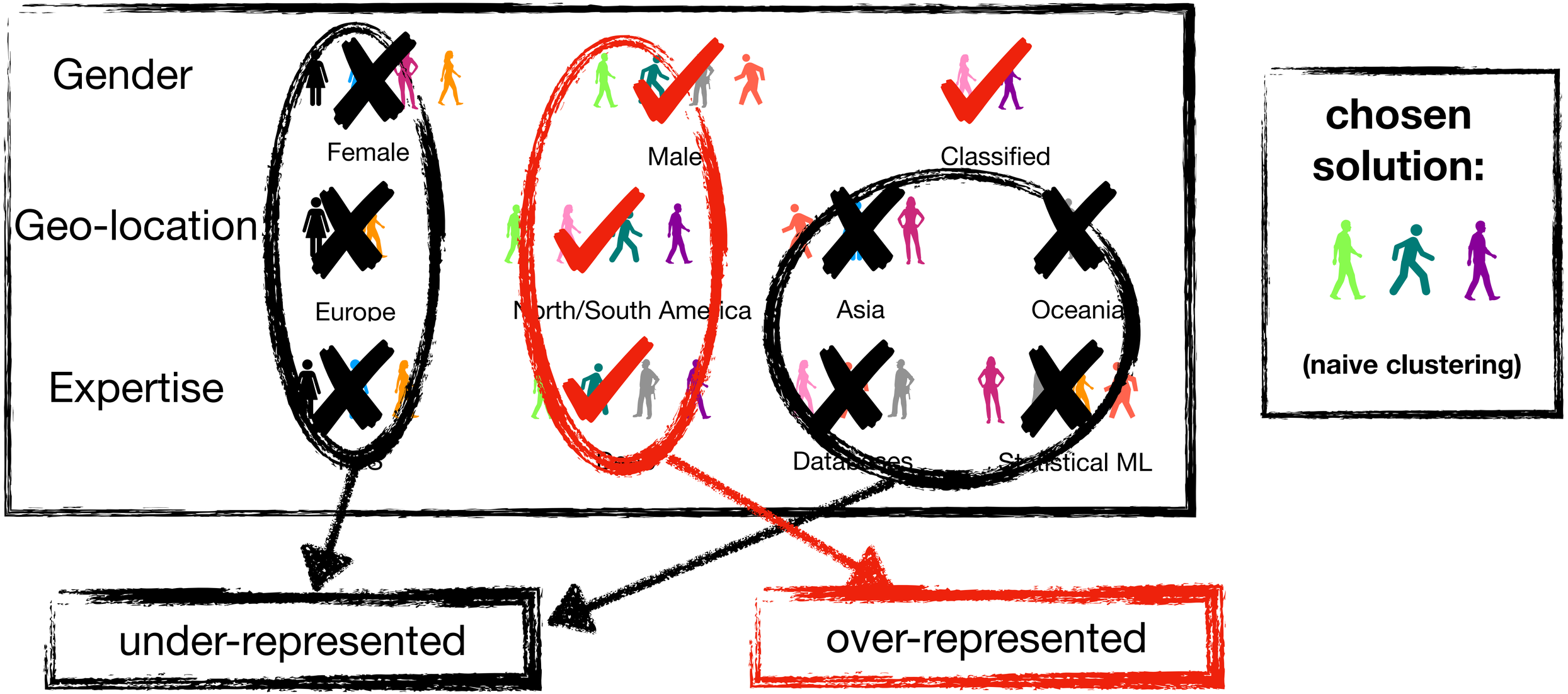
# Motivation

**task:** form a diverse-committee s.t

- experts from different fields
- representation of genders
- different geographical location
- minimise clustering cost, $k$-median/$k$-means clustering objective

Image source: clipartsuggest.com

# Two sides of the problem



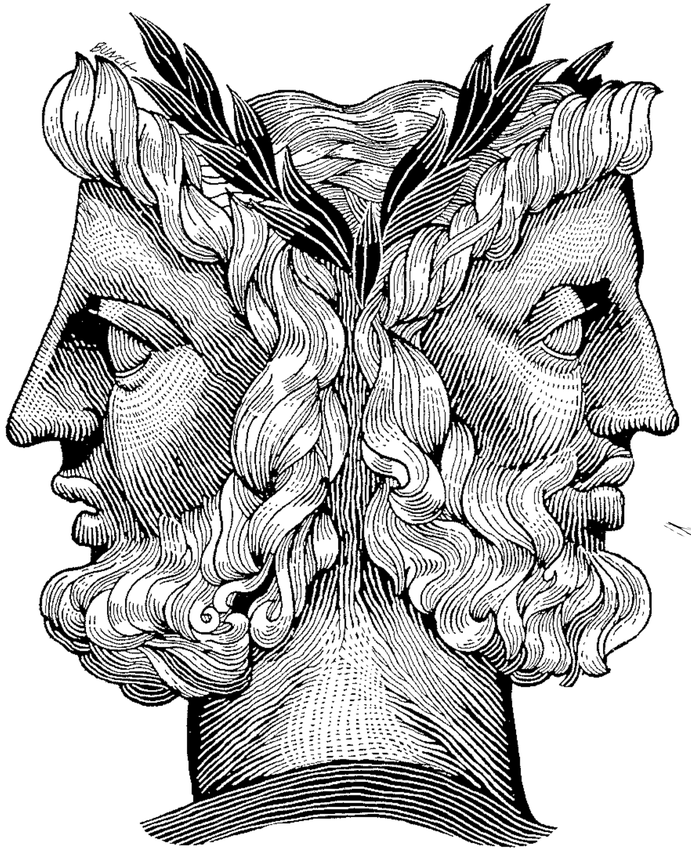
# Two sides of the problem

**Avoid over-representation**

**Avoid under-representation**

Red-blue median problem
Hajiaghayi et al. 2010, 2012

Thejaswi et al. 2021
ECML-PKDD



Matroid-median problem
Krishnaswamy et al. 2010
Chen et al. 2016
Swamy et al. 2018

The problems are mathematically different and lead to problems with different complexity

**Are these problems the same?**



Image source: pinterest.com

# Diversity-aware $k$-median

- given
  a set of clients $C \subseteq U$ and a set of facilities $F \subseteq U$,
  a distance function $d : C \times F \to R_+$,
  a collection $\mathcal{G} = \{G_1, \ldots, G_t\}$ of facility groups $G_i \subseteq F$,
  a set $\vec{r} = \{r_1, \ldots, r_t\}$ of lower-bound thresholds,
  an integer $k \leq |F|$

- find a subset $S \subseteq F$ of facilities such that
  size $|S| = k$,
  constraint $|S \cap G_i| \geq r_i$ is satisfied for all $i \in [t]$,
  minimises the cost function $cost(S) = \sum_{c \in C} \min_{s \in S} d(c, s)$

avoid under-representation
$|S \cap G_i| \geq r_i$

avoid over-representation
$|S \cap G_i| \leq r_i$
red-blue median (Hajiaghayi et al.)
matroid median (Krishnaswamy et al.)

**$k$-median ($k$-means) clustering with lower-bound constraints**

# Hardness and algorithmic results

Thejaswi et al. 2021 ECML-PKDD

| Problem | **NP**-hard | FPT($k$) | Approx. factor | Approx. method |
|---|---|---|---|---|
| Intractable case: intersecting facility groups | | | | |
| General variant | ✓ | ✗ | inapproximable | |
| Tractable cases: disjoint facility groups | | | | |
| $t > 2$, $\sum_{i \in [t]} r_i = k$ | ✓ | open | 8 | LP |
| $t > 2$, $\sum_{i \in [t]} r_i < k$ | ✓ | open | 8 | $\mathcal{O}(k^{t-1})$ calls to LP |
| $t = 2$, $r_1 + r_2 = k$ | ✓ | open | $5 + \epsilon$ | local search |
| $t = 2$, $r_1 + r_2 < k$ | ✓ | open | $5 + \epsilon$ | $\mathcal{O}(k)$ calls to local search |

# Hardness results

Corollary 3.1. Assume SETH. For all $k \geq 3$ and $\epsilon > 0$, there exists no $O(|F|^{k-\epsilon})$ algorithm to solve Div-$k$-Med (Div-$k$-Means resp.). Furthermore, there exists no $O(|F|^{k-\epsilon})$ algorithm to approximate Div-$k$-Med (Div-$k$-Means resp.) to any multiplicative factor.

* brute-force enumeration is the best we can expect for this problem

* approximating the problem to any multiplicative factor is at least as hard as SAT

# Hardness results

Proposition 3.2. For any function $f(k)$, finding $f(k)$ facilities to approximate the Div-$k$-Med (Div-$k$-Means resp.) cost to any multiplicative factor in FPT time with respect to parameter $k$ is W[1]-hard.

* no hope for relaxing the number of facilities by any function of $k$

* even if we relax to $f(k)$ we cannot approximate in $FPT(k)$ time

***** a problem is fixed parameter tractable w.r.t parameter $k$ if there exists an algorithm with runtime of the form $f(k) \cdot poly(|U|)$. Here $f(k)$ can be an exponential or sub-exponential function**

# Hardness results

Proposition 3.3. Finding an optimal solution for Div-$k$-Med (Div-$k$-Means resp.) is W[2]-hard with respect to parameters $(k,t)$, $(r,t)$ and $(\mu,t)$.

* no exact FPT algorithms are possible with respect to many natural parameters of the problem

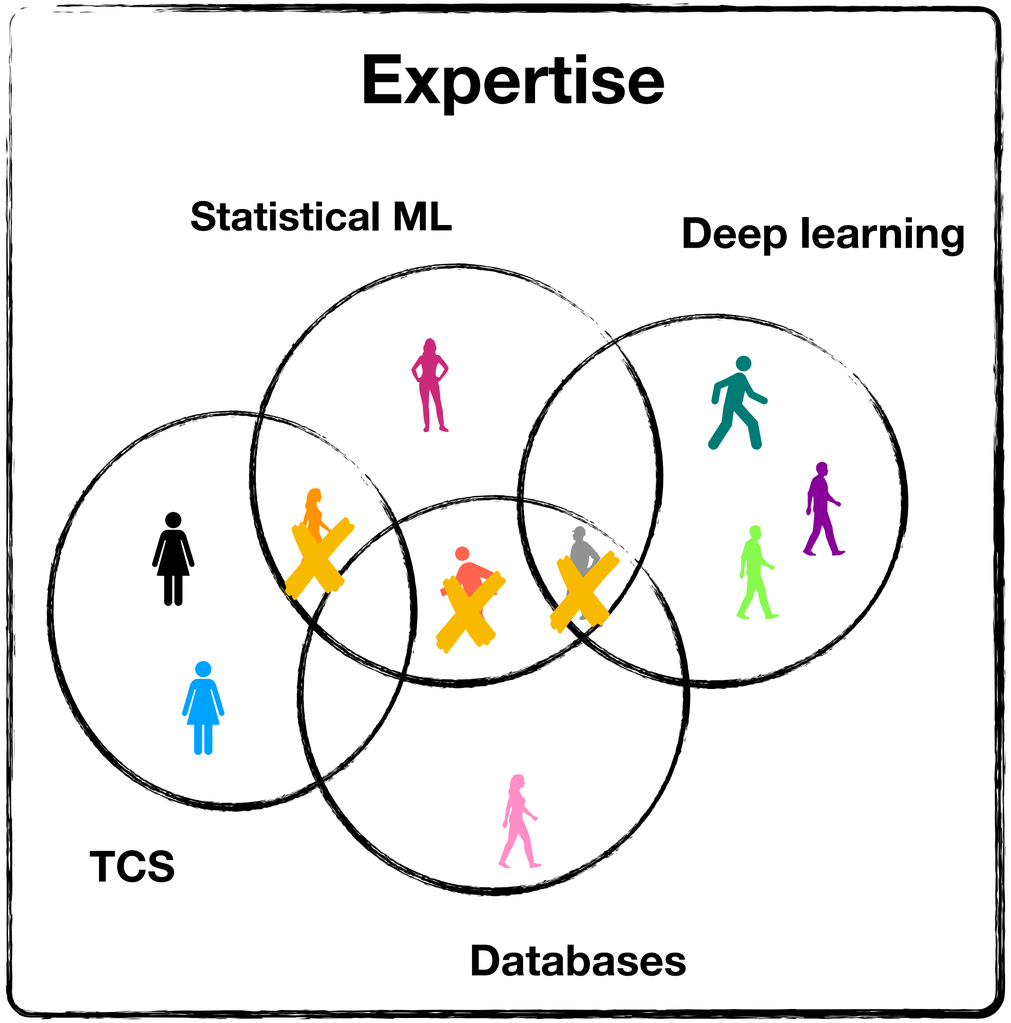
# Algorithmic results

Theorem 4.1. For every $\epsilon > 0$, there exists a randomized $(1 + \frac{2}{e} + \epsilon)$ approximation algorithm for Div-$k$-Med with running time $f(k, t, \epsilon) \cdot \text{poly}(|U|)$, where $f(k, t, \epsilon) = O\left(\left(\frac{2^t k^3 \log^2 k}{\epsilon^2 \log(1+\epsilon)}\right)^k\right)$. The approximation ratio is tight for any FPT algorithm w.r.t $(k,t)$, assuming Gap-ETH.

# Overview of algorithm

1. Find feasible constraint patterns (brute-force enumeration).
2. Create an instance of a $k$-MED-$k$-PM problem for each set of facility types satisfying constraints.
3. Reduce the number of clients via coresets [1].
4. Guess *leaders* from a set of clients in coreset and guess distances of the closest facility in the optimal solution.
5. Making use of recent developments in submodular maximization subject to matroid constraint we obtain a $(1 + \frac{2}{\epsilon} + \epsilon)$ approximation.

# Feasible constraint patterns



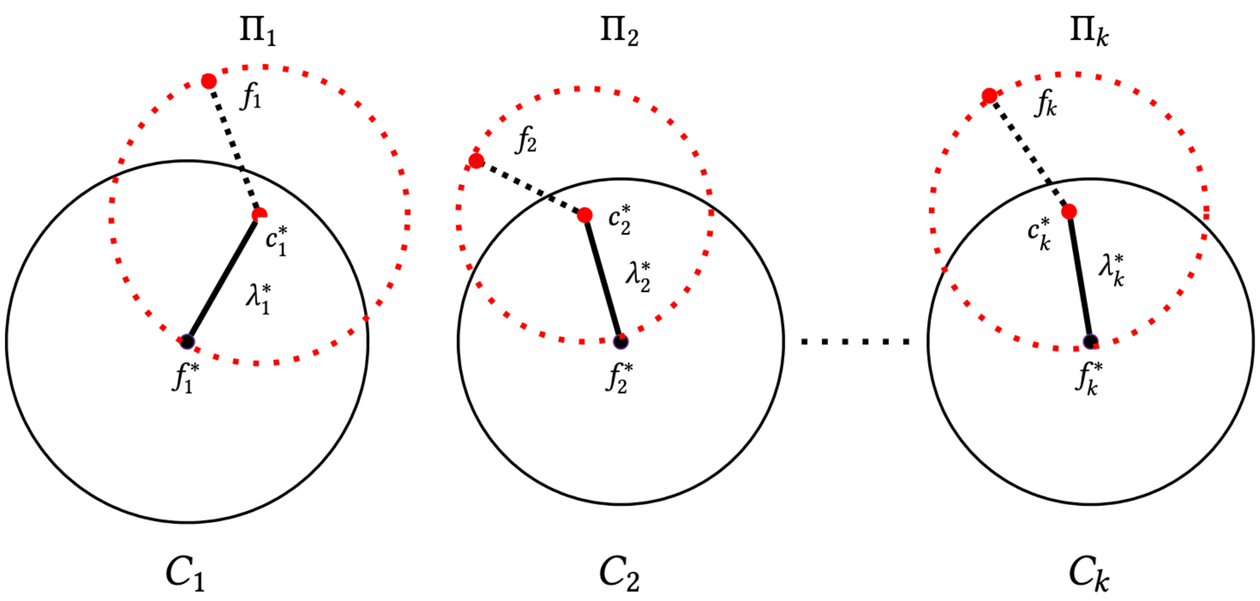

- requirements: $k = 5,\ \vec{r} = \{1,1,1,1\}$
- generate subset lattice of groups
  - intersection of more than one subset forms a group
  - groups from input $\mathcal{G} = \{G_1, \ldots, G_t\}$
- at most $2^t$ groups - subset lattice
- assign values in range $[0,\ldots,k]$ to each group such that the sum of assignment is at most $k$
- $O(2^{tk})$ time - find all feasible assignments
- efficient dynamic program for finding one feasible assignment

# $k$-median $k$-partition matroid

- given
  a set of clients $C \subseteq U$ and a set of facilities $F \subseteq U$,
  a distance function $d : C \times F \to R_+$,
  a partition $\mathscr{E} = \{E_1, \ldots, E_k\}$ of facility groups $E_i \subseteq F$, and $E_i \cap E_j = \emptyset$
  an integer $k \leq |F|$

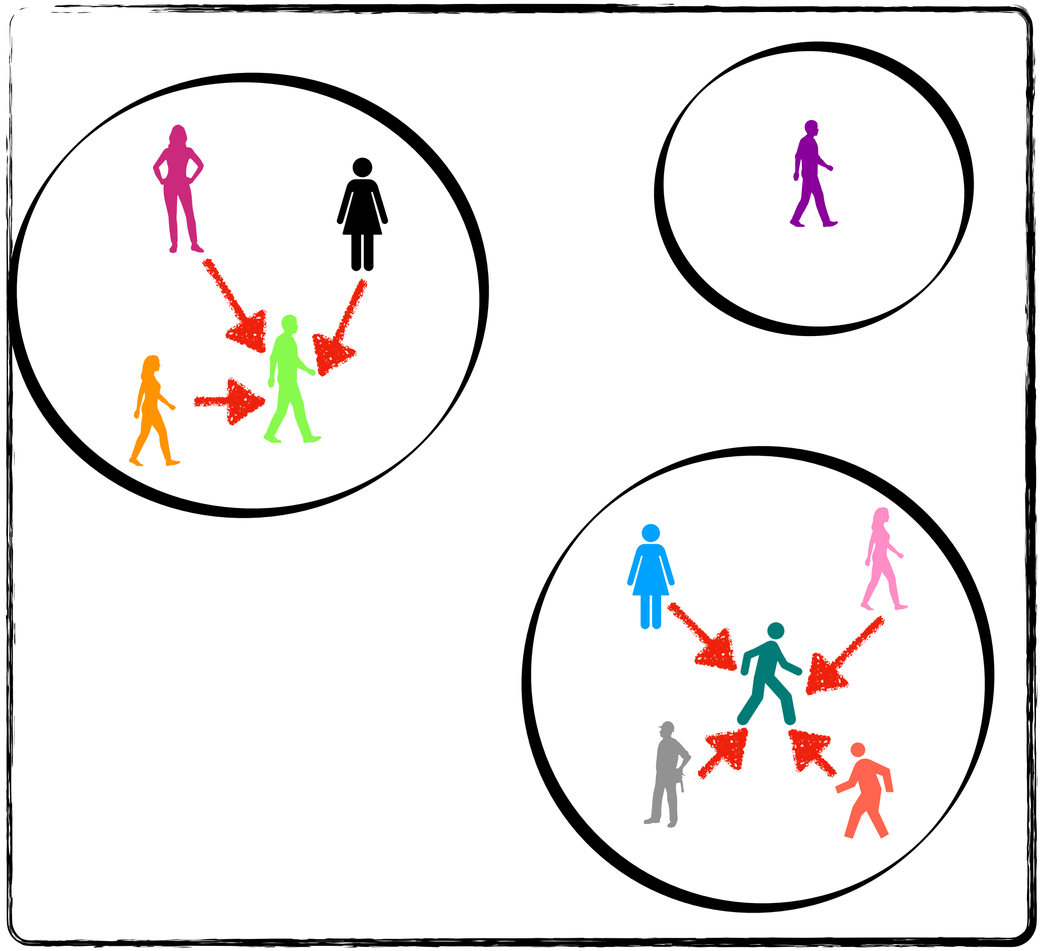
- find a subset $S \subseteq F$ of facilities such that
  size $|S| = k$,
  constraint $|S \cap E_i| \leq 1$ is satisfied for all $i \in [k]$,
  minimises the cost function $cost(S) = \sum_{c \in C} \min_{s \in S} d(c, s)$

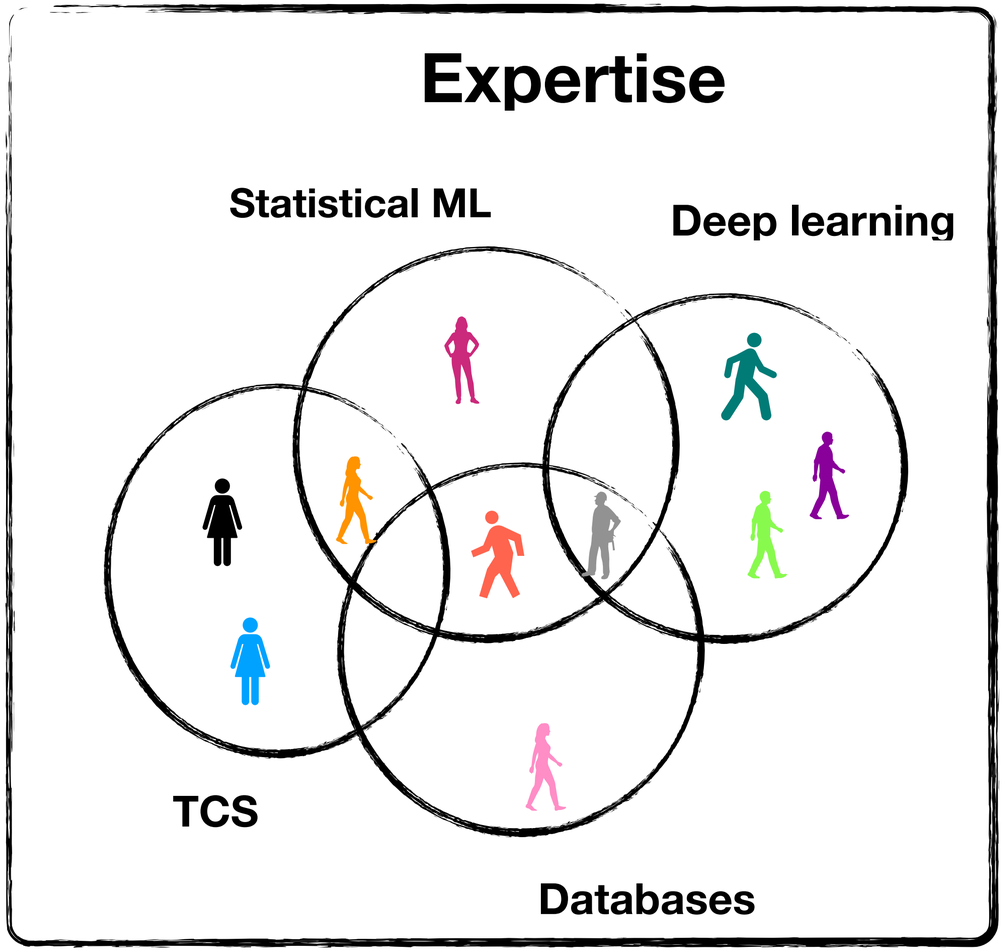
# Approximating $k$-Med-$k$-PM




- $f_1^*, \ldots, f_k^*$ be optimal facilities
- $\lambda_1^*, \ldots, \lambda_k^*$ be distances of nearest clients
- $f_1, \ldots, f_k$ chosen set of facilities
- rerouting gives $3$-approximation
- enumerate all possible $\lambda_1^*, \ldots, \lambda_k^*$
- coreset of size at most $O(k \log k)$
- using sub modularity we can get $1 + \frac{2}{e} + \epsilon$ approximation

# Bicriteria approximation



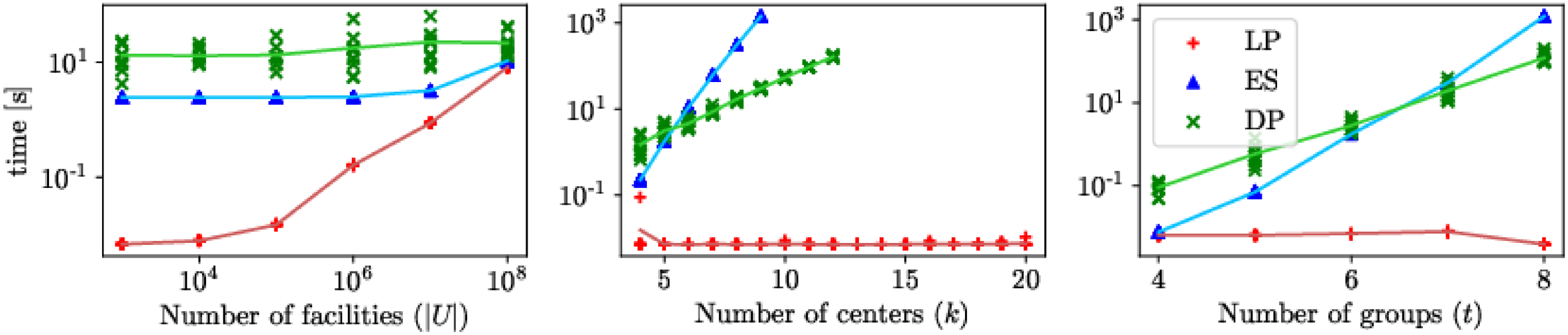



**Solve clustering and feasibility independently**

# Algorithmic results

| Approx. | Approx. factor | Runtime method |
|---|---|---|
| $(3+\epsilon, 2k)$ | LS + LP | $\mathcal{O}^*\left(2^{tk}\right)$ |
| $(3+\epsilon, 2k)$ | LS + DP | $\mathcal{O}^*\left(kt2^t(r+1)^t\right)$ |
| $(1+\frac{2}{e}+\epsilon, k)$ | FPT$(k, t, \epsilon)$ | $\mathcal{O}^*\left(\left(\frac{2^t k^3 \log^2 k}{\epsilon^2 \log(1+\epsilon)}\right)^k\right)$ |

# Experiments — scalability



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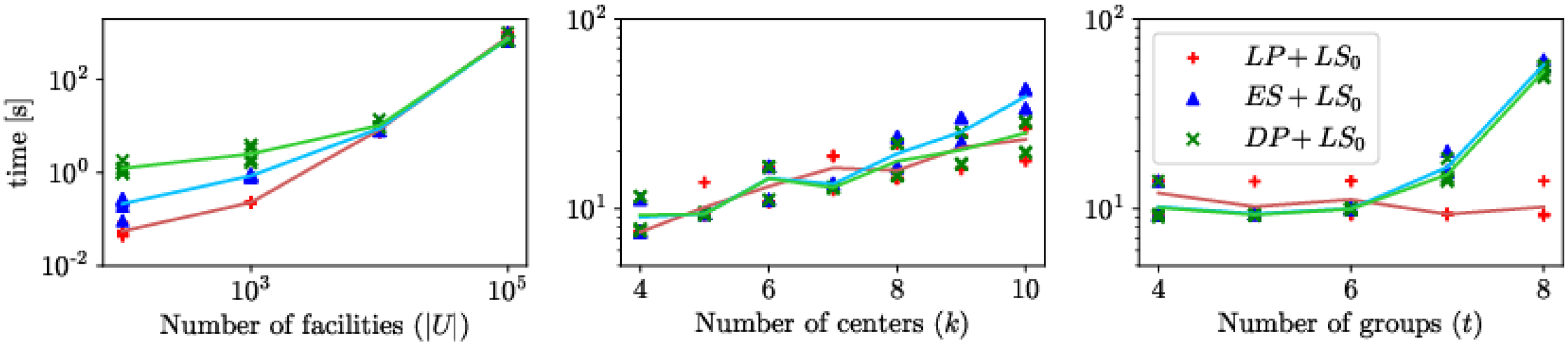

# Conclusions

- diversity-aware clustering is a combination of two fundamental problems
  - satisfying lower-bound constraints
  - finding centres which minimise the clustering objective
- hardness results
  - in-approximable in polynomial time, $FPT(k), FPT(t)$
- algorithmic results
  - $FPT(k, t, \epsilon)$ approximation
  - bicriteria approximation
  - heuristic solutions
- future work
  - $FPT$ algorithm is optimal but not practical
  - it remains open to present practical solution with theoretical guarantees

# Thank you

**source code : github.com/suhastheju/diversity-aware-clustering**